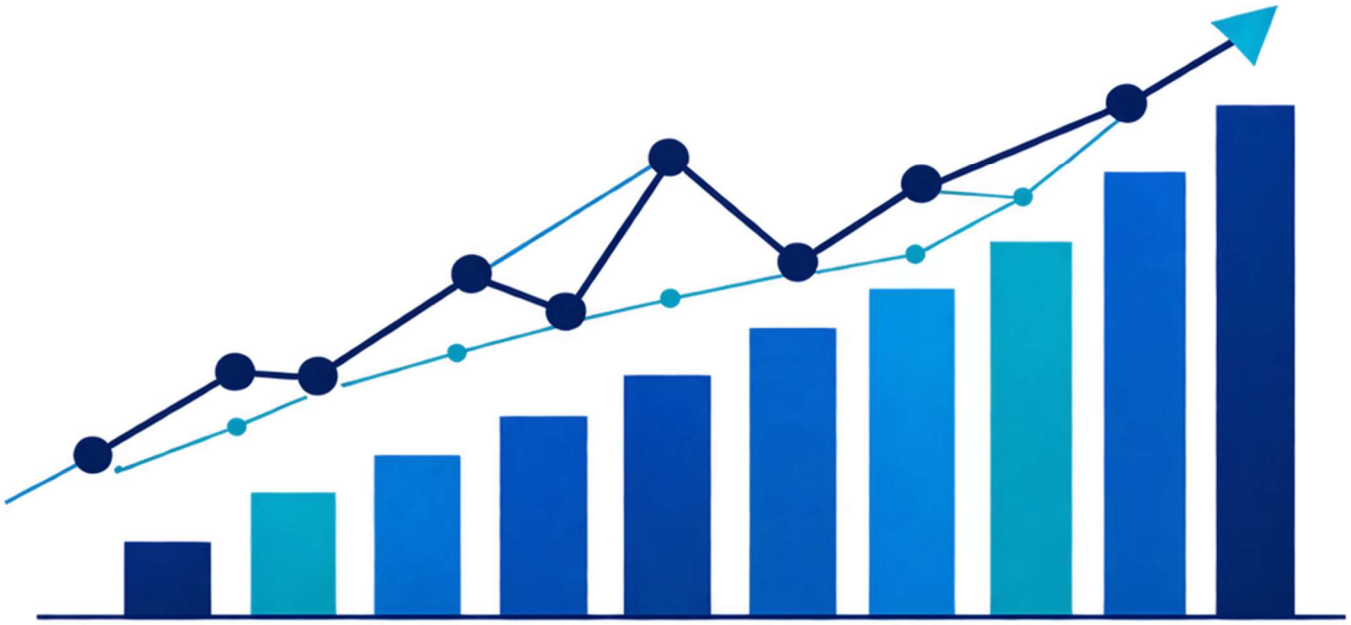
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Business Intelligence & Dashboards For Data Analyst

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ANALYZE BETTER.
GROW FASTER.

Business Intelligence & Dashboards

A Complete 10-Chapter Practice Workbook

First Edition

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Made in India with for Data Analysts Worldwide

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INTERVIEW-BASE UNIT TEST

Unit 1: Dashboard Design & UX Strategy

Business Intelligence & Dashboards

LEVEL 1: WARMUP TEST — 5 MCQ Questions

Instructions: Choose the single best answer for each question.

Q1. Which of the following is a LEAD indicator?

A) Total revenue earned last quarter
B) Number of sales calls made this week
C) Annual profit margin
D) Customer churn rate last month

Q2. What does the "5-Second Rule" in dashboard design state?

A) A dashboard must load in 5 seconds
B) Users should click a chart within 5 seconds
C) The most important insight should be understood within 5 seconds of viewing
D) Tooltips should appear within 5 seconds of hovering

Q3. Which Gestalt Principle states that elements enclosed within a border are perceived as a group?

A) Proximity
B) Similarity
C) Continuity
D) Enclosure

Q4. Which chart type is MOST appropriate to show how revenue changed month by month over a year?

A) Pie Chart
B) Scatter Plot
C) Line Chart
D) Heatmap

Q5. What is the standard format for writing a User Story in dashboard design?

A) "The dashboard should show [metric] for [user]."
B) "As a [type of user], I want to [action], so that I can [goal]."
C) "When [condition], display [chart] to [user]."
D) "Build [feature] for [user] because [reason]."

🎯 LEVEL 2: INTERVIEWER TEST — 15 Intermediate Semi-Advance Questions

Answer all questions in a conversational, thoughtful student format — acknowledge the "why" behind concepts, keep tone practical, and include brief code examples where relevant.

📝 Section A: Descriptive Questions (4 Questions)

Q1. Explain the difference between Lead and Lag Indicators. Why is it important to include BOTH types on a business intelligence dashboard? Provide examples from two different industries.

☑ **Answer**

Q2. Describe all five Gestalt Principles relevant to dashboard design. For each principle, explain how misapplying it can negatively affect user experience.

☑ **Answer**

Q3. What is Color Theory in the context of dashboard design? Explain the three roles of color (Categorical, Sequential, Diverging) with real-world dashboard examples for each.

☑ **Answer**

Q4. Explain the concept of "Request Bloat" in dashboard design. What causes it, and what professional techniques can an analyst use to say no while maintaining stakeholder relationships?

☑ **Answer**

Section B: Scenario-Based Descriptive Questions (4 Questions)

Q1. SCENARIO: A hospital's operations team asks you to build a single dashboard for three users: the Hospital Director (needs bed occupancy % at a glance), Ward Nurses (need today's patient list by ward), and a Health Data Analyst (needs historical admission patterns with filters). How would you approach this, and what interactivity features would you use?

☑ **Answer**

Q2. SCENARIO: You have designed a clean 8-chart sales dashboard. A senior VP storms into your meeting and demands you add 7 more charts immediately — including a map of revenue by country for all 50 countries, hourly revenue fluctuations, and a chart showing revenue in 23 different currencies. How do you respond?

☑ **Answer**

Q3. SCENARIO: An e-commerce analyst is building a campaign performance dashboard. The marketing team wants to compare 5 campaigns' total clicks, show how daily website traffic changed over 30 days, AND show what % of revenue came from each of 4 product categories. Which chart types would you use for each, and why?

☑ **Answer**

Q4. SCENARIO: You are reviewing a colleague's dashboard and notice: a 3D pie chart with 9 slices, a dual-axis chart mixing revenue and profit %, and a bar chart with a Y-axis starting at 850,000 instead of 0. Identify all design violations, explain why each is problematic, and propose the correct solution.

☑ **Answer**

💻 Section C: Coding Questions (4 Questions)

Q1. Using Python and pandas, write code to calculate both a Lead KPI (7-day rolling average of daily active users) and a Lag KPI (monthly total revenue). The dataset has columns: date, daily_active_users, and revenue.

☑ **Answer**

Understanding the "why" here matters: rolling averages smooth out daily noise so you can see the trend signal clearly — that's what makes it useful as a lead indicator. Monthly total is a clean lag confirmation.

Code:

☑ **Answer**

Q2. Write a Python function that evaluates a KPI value against a target and returns a color-coded RAG (Red/Amber/Green) status. The function should accept: value, target, and an amber_threshold (default 10%). Apply this to a list of 5 KPIs and print a styled summary.

☑ **Answer**

The RAG pattern is foundational to dashboard color theory — it implements the diverging color role (green = above, amber = warning, red = below). The threshold logic is where the real value is.

Code:

☑ **Answer**

Q3. Using Python and matplotlib, write code to create a properly designed dashboard chart panel that demonstrates chart selection best practices: a bar chart for comparison, a line chart for trends, and a donut chart for composition — all in one figure with correct formatting.

☑ **Answer**

Chart selection best practice says: match the chart to the relationship being shown. This code demonstrates all three categories in one figure.

Code:

☑ **Answer**

Q4. Write a Python function called apply_5_second_rule() that takes a list of KPI dictionaries (with name, value, target, and priority fields) and sorts them by priority, flags any that fail a simple threshold test, and returns a formatted layout plan showing Zone 1 (top-left), Zone 2 (top-right), and Zone 3 (bottom) assignments.

☑ **Answer**

This is the 5-Second Rule and Information Hierarchy implemented in code — the most critical KPI gets Zone 1 automatically based on priority and performance status.

Code:

☑ **Answer**

Section D: Database Scenario-Based Questions (3 Questions)

Q1. DATABASE SCENARIO: You have a sales_transactions table with columns: transaction_id, salesperson_id, region, sale_date, revenue, and target_revenue. Write SQL to generate a KPI summary report showing: total actual revenue, total target revenue, attainment %, and RAG status (Green ≥ 100%, Amber ≥ 90%, Red < 90%) — grouped by region. Sort by attainment ascending (worst-performing regions first).

☑ **Answer**

This is a classic dashboard backend query — grouping real vs. target data and applying business logic (RAG classification) inside SQL. Sorting worst-first is intentional: it ensures problem regions are visible immediately on the dashboard.

SQL:

☑ **Answer**

Q2. DATABASE SCENARIO: You are building an interactivity feature — a drill-down from monthly revenue to weekly breakdown. The dashboard_sales table has: sale_id, sale_date, region, product_category, revenue. Write two SQL queries: Query 1 returns the monthly revenue summary. Query 2 accepts a month parameter and returns weekly revenue for that month (as a drill-down result).

☑ **Answer**

This is the SQL backbone of a drill-down interaction — the first query populates the summary chart, and clicking a bar triggers the second query with the selected month as a parameter. Understanding this connection is key to building interactive dashboards.

Query 1 — Monthly Revenue Summary (Summary Layer):

☑ Answer

Query 2 — Weekly Drill-Down (accepts month parameter e.g. '2024-03'):

☑ Answer

Q3. DATABASE SCENARIO: To prevent request bloat, you are asked to build a "dashboard usage analytics" table to track which charts are actually being viewed. Design the schema for a chart_view_log table and write a SQL query that identifies "zombie charts" — charts that have had zero views in the last 30 days — so they can be removed from the dashboard.

☑ **Answer**

This is a meta-analytics problem — using data to make decisions ABOUT the dashboard itself. Finding zombie charts is the evidence-based approach to fighting request bloat without needing subjective debates with stakeholders.

Schema Design:

☑ Answer



Zombie Chart Detection Query:

